Introduction to Microsoft Excel

1. What Is Microsoft Excel?

Microsoft Excel is a **spreadsheet** program. A spreadsheet is a big table made of rows and columns where you can store data and do calculations. If you change one number, Excel updates the results for you, which is why Data Analysts use it every day.

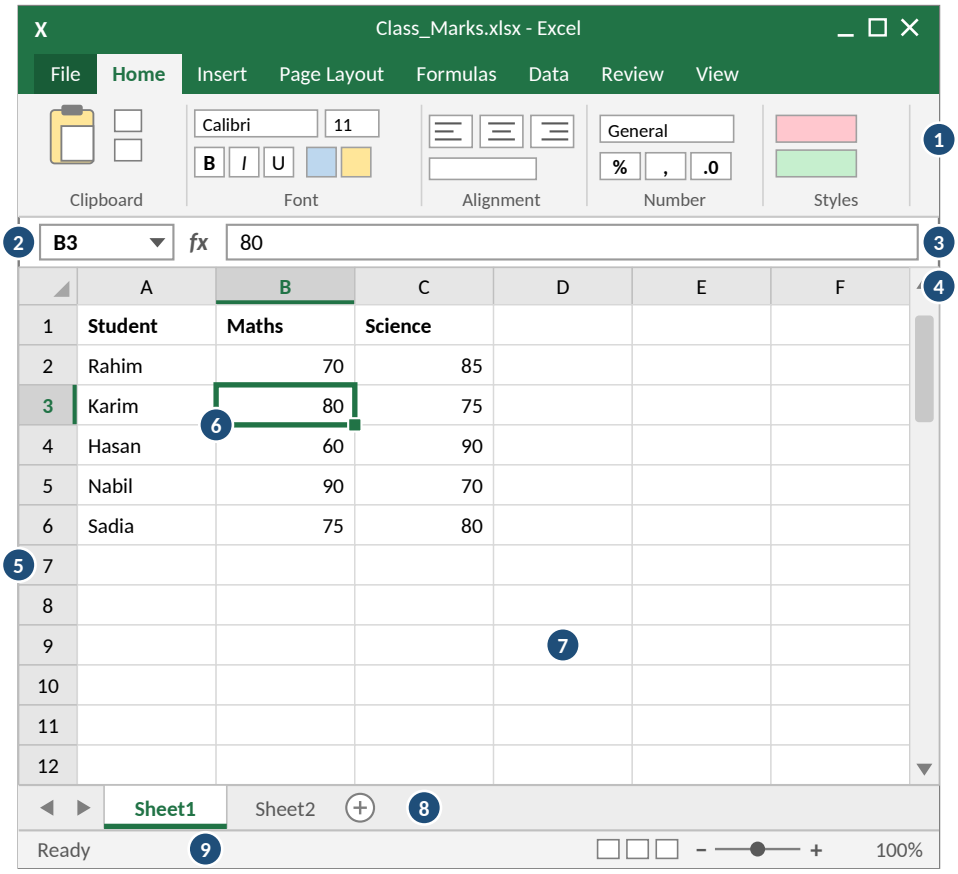
- **A+ Student marks** — grades & rankings
- **Sales** — revenue by product
- **$ Budgeting** — income & expenses
- **Inventory** — stock levels
- **Data analysis** — trends & insights

Raw Data (scattered & unsorted) → **Organized Data** (clean rows & columns) → **Analysis** (calculate & summarize) → **Report** (charts & results to share)

2. A Brief History of Excel

- **1985** — **Excel for Macintosh**: The first version of Excel
- **1987** — **Excel for Windows**: Brings Excel to PC users
- **2011–2020** — **Microsoft 365**: Cloud storage & teamwork
- **Today** — **Modern data tools**: Charts, tables & smart tools

3. Understanding the Excel Interface



	A	B	C	D	E	F
1	Student	Maths	Science			
2	Rahim	70	85			
3	Karim	80	75			
4	Hasan	60	90			
5	Nabil	90	70			
6	Sadia	75	80			
7						
8						
9						
10						
11						
12						

1. **Ribbon & tabs** — Commands grouped by task: Home, Insert, Data, View...
2. **Name Box** — Shows the address of the selected cell (here, B3).
3. **Formula Bar** — Shows and edits the content of the selected cell.
4. **Column headings** — Letters A, B, C... name each column.
5. **Row headings** — Numbers 1, 2, 3... name each row.
6. **Active cell** — The selected cell, outlined in green. Typing goes here.
7. **Worksheet grid** — The area of cells where you enter data.
8. **Sheet tabs** — Click a tab to switch between worksheets.
9. **Status bar** — Shows Ready/Edit mode, zoom, and quick information.

4. Workbook vs. Worksheet

A **workbook** is the whole Excel file (.xlsx). A **worksheet** is a single page, or tab, inside it. Like a notebook with many pages, one workbook can hold many worksheets.

Term	What it is
Workbook	The file: Class_Marks.xlsx
Worksheet	One tab inside it: Term 1

WORKBOOK Class_Marks.xlsx

	A	B	C	D
1	Student	Maths	Science	
2	Rahim	70	85	
3	Karim	80	75	
4	Hasan	60	90	

WORKSHEET — One page of data: a grid of rows and columns

Term 1 | Term 2 | Summary ← Worksheet tabs

5. Rows, Columns, Cells, and Ranges

COLUMN
vertical, lettered A, B, C...

RANGE A1:C4
a block of cells: first cell : last cell

	A	B	C	D	E
1	**Student**	**Maths**	**Science**		
2	Rahim	70	85		
3	Karim	80	75		
4	Hasan	60	90		
5	Nabil	90	70		
6	Sadia	75	80		

ROW
horizontal, numbered

TIP
Click any cell and its address appears in the **Name Box**.

B + 3 = B3

CELL: where a row and a column meet. Its **address** is the column letter + row number.

6. Entering and Formatting Data

BEFORE *messy & inconsistent*

student	marks	date	score %
rahim	70	5/3/26	0.7
KARIM	80	March 5	80
hasan	sixty	05-03-2026	60%
Nabil	90	5 Mar	0.9

Mixed case • number stored as text • different date styles • mixed % styles

Format & clean

AFTER *clean & consistent*

Student	Marks	Date	Score %
Rahim	70	05-Mar-2026	70%
Karim	80	05-Mar-2026	80%
Hasan	60	05-Mar-2026	60%
Nabil	90	05-Mar-2026	90%

Consistent names • real numbers • one date style • one % style

- **Type:** click a cell, type, and press **Enter**. To change it later, double-click the cell.
- **Text** (like names) lines up on the left. **Numbers, dates, and percentages** line up on the right.
- **Make it neat:** bold the header row, add borders, and widen columns so everything is visible.
- **Stay consistent:** use one style in each column so Excel can sort and calculate correctly.

7. Good Habits for Clean Data

GOOD

Student	Maths	Science
Rahim	70	85
Karim	80	75
Hasan	60	90

Header on top • no gaps • one value per cell

AVOID

Class Marks 2026		
(blank row)		
Student	Maths	Science
Rahim	70 marks	85
Karim	80	75

Merged title • blank row • text mixed with numbers

HABITS

- Start your table in cell **A1**.
- Give every column a clear **header**.
- Keep **one table** on each sheet.
- Save often with **Ctrl + S**.
- Fix mistakes with **Ctrl + Z** (undo).

8. Preparing for Formulas

A **formula** is an instruction that tells Excel to calculate something. It always starts with an equals sign (**=**); for example, `=2+3` shows 5. A **function** is a ready-made formula with a name, built into Excel to save you time.

A1 | *fx* | =2+3

press Enter

5 ← the cell shows the result

"Now that you understand Excel's workspace and how data is organized, you are ready to learn Excel formulas and functions."